

A Major Project Report on

EXPLORING THE POTENTIAL OF ALOE VERA
FIBER COMPOSITE MATERIALS

*Submitted to Bihar Engineering University (BEU), Patna in partial
fulfilment of the requirement for the award of the*

Bachelor of Technology
In
Mechanical Engineering
by

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Under the Supervision of

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August 2024

Candidate's Declaration

This is to declare that we have carried out the project work “**Exploring the Potential of Aloe Vera Fiber Composite Materials**” in the Department of Mechanical Engineering, Government Engineering College Kishanganj for the partial fulfillment of the award of the degree of Bachelor of Technology. The work presented in this report is authentic to the best of our knowledge and has been done from April 2024 to June 2024.

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Certificate

This is to certify that Mr. Avinash Kumar (21102142908) and Mr. Kulesh Gaurav (20102142001) have completed their Major Project on “**Exploring the Potential of Aloe Vera Fiber Composite Materials**” in the partial fulfillment of the award of the Degree of Bachelor of Technology in Mechanical Engineering. They have worked under the supervision of Prof. Vikash Kumar, Asst. Professor, Dept. of Mechanical Engineering, and Co-Supervision of Prof. Dr. Md. Meraj, Dept. of Mechanical engineering, Government Engineering College Kishanganj, Kishanganj.

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Acknowledgement

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Abstract

This project focuses on the creation and evaluation of a composite material made from Aloe vera fibers, combined with epoxy resin and hardener. Aloe vera fibers, known for their natural strength, lightweight properties, and environmental friendliness, were chosen as the reinforcing material. The epoxy resin serves as the binding matrix, while the hardener facilitates the curing process to form a solid, durable composite. The primary objective of this project is to explore the mechanical properties of the resulting composite, including tensile strength, flexural strength, and impact resistance, and to determine how these properties vary with different fiber orientations and resin-to-hardener ratios. The project also aims to assess the potential applications of this sustainable composite material in various industries, such as automotive, construction, and consumer goods. Through this research, Aloe vera fiber-reinforced epoxy composites are positioned as an eco-friendly alternative to conventional synthetic composites, offering both environmental and performance benefits.

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Nomenclature

ACRONYMS	DESCRIPTION
NFRCS	Natural fiber reinforcement composites
ASTM	American Standard of Testing and Materials
PMCs	Polymer Matrix Composites
MMCs	Metal Matrix Composites
CMCs	Ceramic Matrix Composites
UTM	Universal Testing Machine
Wt. %	Weight percentage
t	Time
T	Temperature
° C	Degree Centigrade
MPa	Mega Pascal
GPa	Giga Pascal
Wt%	Weight Percentage
R&D	Research and Development

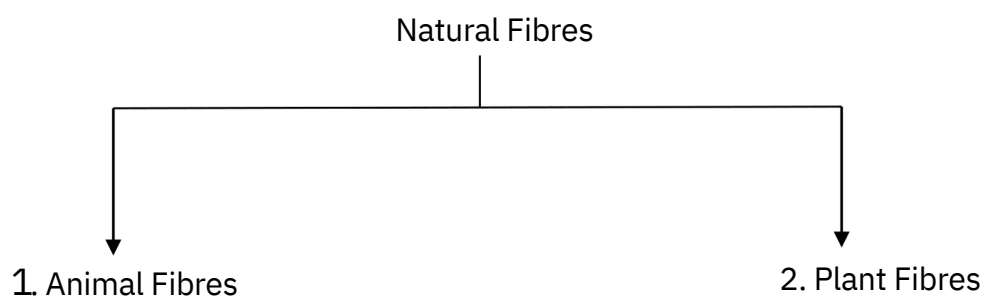
CHAPTER 1

Introduction

1.0 Overview of Natural fibre Composite

Natural fibres are fibres that are produced by geological processes, or from the bodies of plants or animals. They can be used as a component of composite materials, where the orientation of fibres impacts the properties. Natural fibres can also be matted into sheets to make paper or felt. Basically, two different types of fibre reinforcement, i.e., naturally occurred and man-made fibre. Glass, carbon and aramid are man-made fibres, also known as synthetic fibre. These fibres have better mechanical properties but they are expensive and not environmentally friendly. Natural fibres are available naturally in abundance quantity. They are obtained from plant, animal and rock, Jute, flax, kenaf, hemp, sisal, bamboo and aloe Vera are common example of plant fibres. These fibres are extracted from different parts of plant like stem, leaf, and seed. The extraction of fibres involves a number of chemical and physical processes. Cellulose, hemicellulose, lignin and pectin are the major components of the plant fibre. Among all the four components, cellulose provides structural strength to fibre. Natural fibres are good sweat absorbents and can be found in a variety of textures. Cotton fibres made from the cotton plant, for example, produce fabrics that are light in weight, soft in texture, and which can be made in various sizes and colours. Clothes made of natural fibres such as cotton are often preferred over clothing made of synthetic fibres by people living in hot and humid climates.

Natural fibre is classified on the basis of origin of source-



1.1 Animal Fibres

Animal fibres are naturally occurring fibres that are sourced from animals, as the name suggests. Animal fibres are made up of different types of proteins in general. Some of the most common animal fibres are wool and silk. You must know that all

animal fibres are extracted from different animals and hence have different properties from one another. Also, the kind of fibres depends from one animal to another.



Figure 1 Animal Fibre [1]

1.2 Plant Fibres

Plant fibres are generally composed of cellulose, often in combination with other components such as lignin. Examples include cotton, hemp, jute, flax, ramie, sisal, and biases. Plant fibres are classified according to their source in plants.

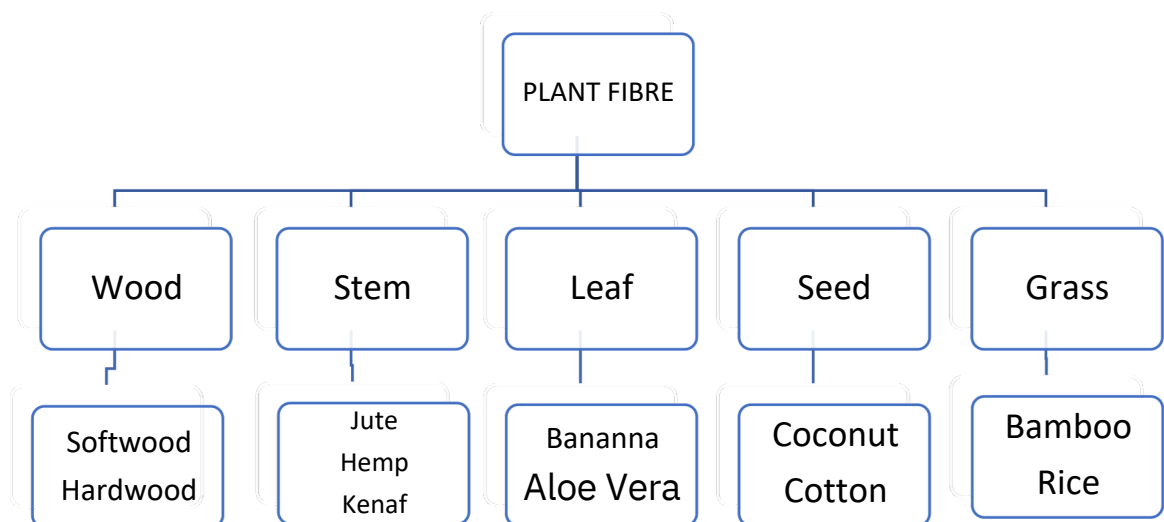


Figure 2 Plant Fibre [2]

Table 1 Composition of few commonly used natural fibres

Fibre	Cellulose (wt.%)	Hemicelluloses (wt.%)	Lignin (wt.%)	Pectin (wt.%)	Moisture (wt.%)	Waxes
Cotton	85-90	85-90	-	0-1	7.85-8.5	0.6
Bamboo	60.8	0.5	32	-	-	-
Jute	61.1-71.5	13.6-20.4	12-13	0.2	12.5-13.7	0.5
Ramie	68.6-76.2	13.1-16.7	0.6-0.7	1.9	7.5-17	0.3
Sisal	66-78	10-14	10-14	10	10-22	2
Aloe Vera	63-64	19	5	-	10-12	-

1.2.1 Types of Plant Fibre



1.2.2 Aloe Vera Fibre

Introduction

The Aloe Vera plant has been known and used for centuries for its health, beauty, medicinal and skin care properties. The name Aloe Vera derives from the Arabic word “Aloe” meaning “shining bitter substance,” while “Vera” in Latin means “true.” 2000 years ago, the Greek scientists regarded Aloe Vera as the universal panacea. The

Egyptians called Aloe “the plant of immortality.” Today, the Aloe Vera plant has been used for various purposes in dermatology.

History Aloe Vera has been used for medicinal purposes in several cultures for millennia: Greece, Egypt, India, Mexico, Japan and China.¹ Egyptian queens Nefertiti and Cleopatra used it as part of their regular beauty regimes. Alexander the Great, and Christopher Columbus used it to treat soldiers’ wounds. The first reference to Aloe Vera in English was a translation by John Good yew in A.D. 1655 of Discords’ Medical treatise *De Material Medica*.² By the early 1800s, Aloe Vera was in use as a laxative in the United States, but in the mid-1930s, a turning point occurred when it was successfully used to treat chronic and severe radiation dermatitis.



Figure 3 Aloe Vera [3]

1.2.3 Mechanical properties of aloe vera fibers:

Property	Value Range	Description
Tensile Strength	50 - 150 MPa	Measures the maximum stress the fibers can withstand while being stretched or pulled before breaking.
Young's Modulus	1 - 5 GPa	Indicates the stiffness of the fibers; higher values mean the fibers are stiffer.
Density	1.3 - 1.5 g/cm ³	Reflects the mass per unit volume of the fibers, indicating they are relatively lightweight.
Elongation at Break	2 - 5%	Describes the fibers' ability to stretch before breaking. Higher values indicate more flexibility.
Moisture Absorption	High (Qualitative)	The fibers can absorb a considerable amount of moisture, which affects their performance in humid conditions.
Thermal Stability	200 - 250°C (Degradation starts)	The temperature range at which the fibers begin to degrade thermally.

1.3. Composite

A composite material is composed of at least two materials, which combine to give properties superior to those of the individual constituents. For our website we refer to fibre reinforced polymer (FRP) composites, usually with carbon, glass, aramid, polymer or natural fibres embedded in a polymer matrix. Other matrix materials can be used and composites may also contain fillers or Nano- materials such as graphene.

The many component materials and different processes that can be used make composites extremely versatile and efficient. They typically result in lighter, stronger, more durable solutions compared to traditional materials.

1.3.1.Objective of composite

The primary reason composite materials are chosen for components is because of weight saving for its relative stiffness and strength. For example, carbon-fiber reinforced composite can be five times stronger than 1020 grade steel while having

only one fifth of the weight. Aluminum (6061 grade) is much nearer in weight to carbon-fiber composite, though still somewhat heavier, but the composite can have twice the modulus and up to seven times the strength.

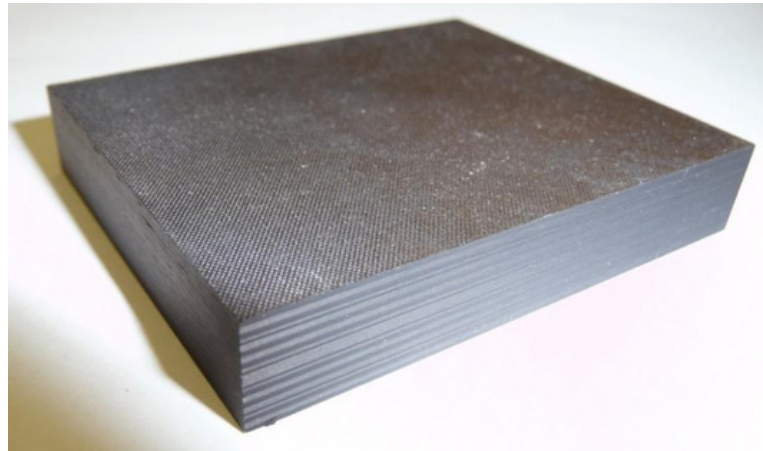


Figure 4 Composite Material [4]

1.3.2.Constituents of Composite

Composite materials are created from individual materials. These individual materials are known as constituent materials, and there are two main categories of it. One is the matrix (binder) and the other reinforcement.

1.3.2.1Matrix

- a) Material that holds the reinforcement in place.
- b) Provides bulk form of the parts or products made of the composite material.
- c) When load is applied the matrix shares the load with the reinforcing element.

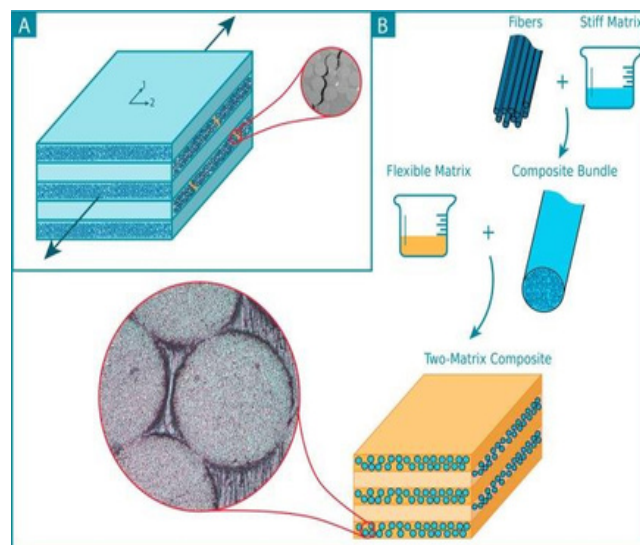


Figure 5 Matrix [5]

1.3.2. 2. Reinforcement

Material that provides the strength to the matrix material.



Figure 6 Reinforcement [6]

1.4 Primary Process

NFRCs can be developed using thermoset as well as thermoplastic material. Thermoplastic contain linear or branched polymeric chain. Therefore, it has tendency to soften after heating and return to its original state after cooling. Whereas, thermosets composed of crosslink and network type polymeric chain. It requires heat for curing. Once thermoset cured, it cannot achieve its initial state. Therefore, primary processing of both types of polymers are different. Thermoset based NFRCs can be processed with hand lay-up, resin transfer molding, vacuum bag molding, pultrusion, filament winding and compression molding. Hand lay-up process can be used for any form of fiber like short furlong fiber, particulate type fiber, unidirectional fiber, bidirectional fiber a woven type fiber.

1.4.1 Thermoplastics

Thermoplastics are defined as polymers that can be melted and recast almost indefinitely. They are molten when heated and harden upon cooling. When frozen, however, a thermoplastic becomes glass-like and subject to fracture. As a result, thermoplastics are mechanically recyclable.

1.4.2 Thermosets

Thermosets composed of crosslink and network type polymeric chain. It requires heat for curing. Once thermo set cured, it cannot achieve its initial state.

Therefore, primary processing of both types of polymers are different.

1.5 Secondary Process

These processes are used to further modify the output of primary manufacturing processes in order to improve the material properties, surface quality, surface integrity, appearance and dimensional tolerance. The input material for these processes must have some specific shape and size; otherwise products may not be accommodated within the machine facility. Various secondary manufacturing processes are Machining-lathe, shaper Assembly- welding, rivet, bolted, brazing....etc.

1.5.1 Need of joining

- a) To fabricate large and intricate composite products.
- b) To create relative motion between mating parts.
- c) For proper load distribution and transfer.
- d) To provide maintenance, repair and service.

1.6 Joining techniques of composites

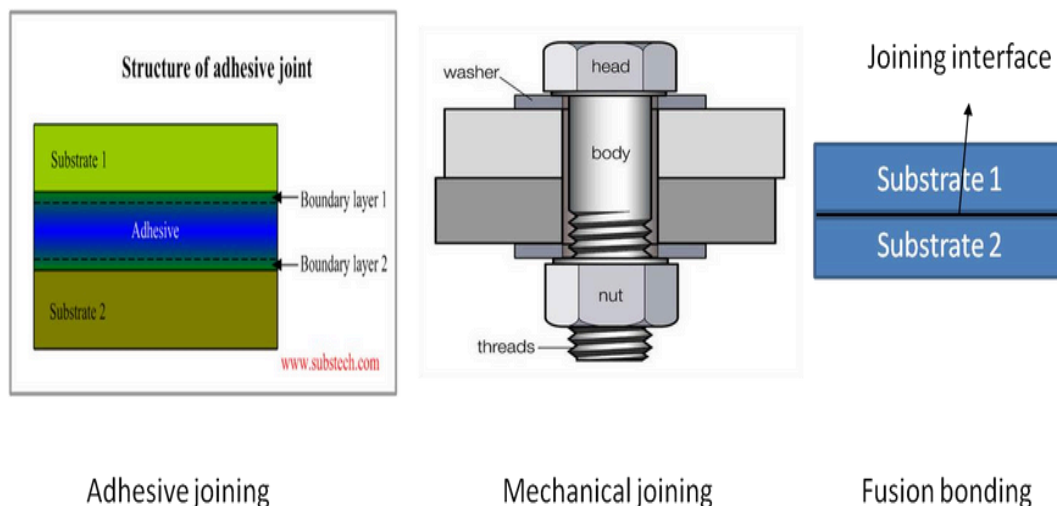


Figure 6.1 Joints technique of composite

1.6.1 Adhesive bonding

- a) A wide variety of materials is available when adhesives are used to bond materials together.

- b) The choice of which adhesive is best is usually dictated by the type of composite to be bonded, the application of the bonded composite, the service environment, and cost.
- c) The most common polymers in the structural adhesives class are: epoxies, polyurethanes, acrylics, cyanoacrylates, silicones, and phenolics.

1.6.1.1 Failure mechanism of adhesive bonding

Failure of adhesive joint can be predicted by three mode, adhesive failure, cohesive failure and mixed (adhesive and cohesive) failure. Showing all the three modes of failure. The failure in adhesive joint can be visualized at the adhesive adhered joint interface. Adhesive failure occurs due to improper surface preparation, wrong adhesive selection and high peel stress. An improper surface preparation reduces the mechanical interlocking between adhesive and adherent which may result in adhesive failure. Wrong adhesive selection again reduces the adhesion between adhesive and adhered which causes adhesive mode of failure. In cohesive mode of failure, either adhered fail or adhesive fail. It is due to high bond strength between adhesive and adhered than their individual strength. In mixed type of failure, adhesive and cohesive failure takes place simultaneously.

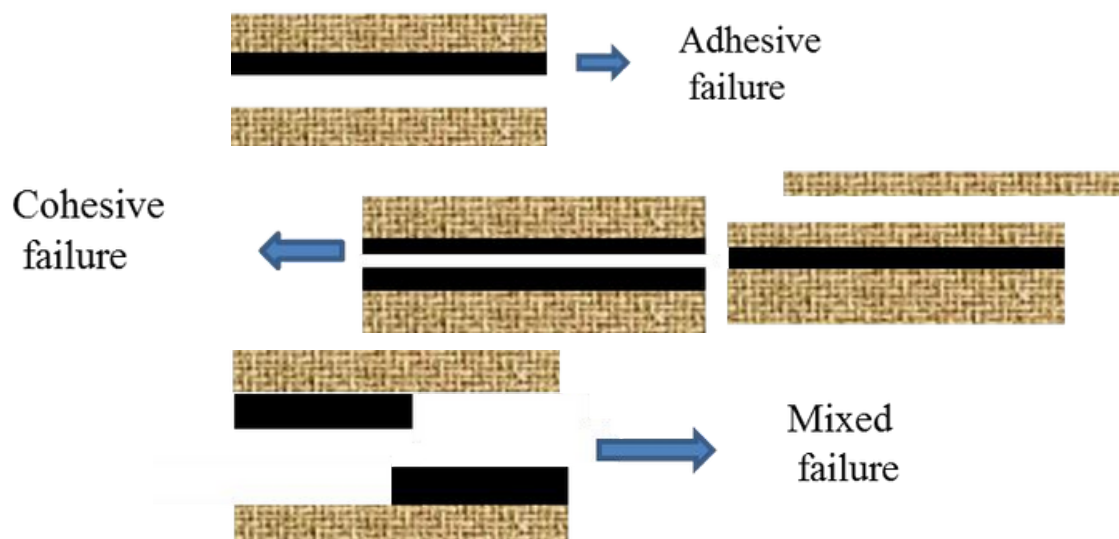


Fig. 7 Failure mechanism in adhesive joint

1.6.2 Fusion bonding

Only thermoplastic based composites may join by using fusion bonding techniques. It is not applicable for thermoset based composites because thermosets are irreversible in nature. After curing, it cannot transform into its initial state. But thermoplastics can change its state after solidification. When the thermoplastics

are heated, it starts softening and after removal of heat, it gets solidified. There is large variety of fusion bonding techniques. It includes: friction welding, thermal welding and electromagnetic welding. Ultrasonic welding, vibration welding and stir welding comes under friction welding; hot plate welding, laser welding and infrared welding comes under thermal welding; and resistance welding, microwave welding and induction welding comes under electromagnetic welding.

1.6.2.1 Factors influencing failure of fusion bonded joint

- a) Heating duration.
- b) Temperature at joint interface.
- c) Welding pressure
- d) Holding time.
- e) Material properties of thermoplastic.
- f) Susceptor (implants) material properties.

1.6.3 Mechanical Joining

Mechanical joining is one of the oldest methods for joining of different kind of materials. It is also frequently used for joining of composites. Thermosets and thermoplastics matrix composites both can be joined by mechanical joining. Among all type of mechanical joining, mechanical fastening is most common technique for mechanical joining. It provides easy assembly and disassembly of components. Therefore, maintenance of joint is very easy in case of mechanical fastening. There is one major defect of mechanical fastening is drilling hole on joining component for fixing the fastener. Presence of hole increases the stress concentration near joining interface and further result for failure of joint. In case composite, drilling a hole is more challenging than metals.

1.6.3.1 Rivets

A riveted joint is a permanent joint with mainly two components (parts to be joined) which are held together by a rivet with the head at top and tail at the bottom.

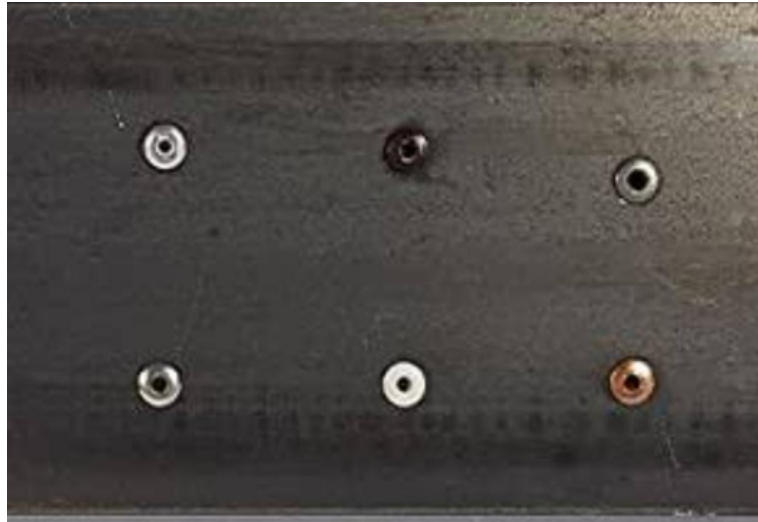


Figure 8 Rivet [7]

1.6.3.2 Bolts and Screws

Bolted joints are one of the most common elements in construction and machine design. They consist of fasteners that capture and join other parts, and are secured with the mating of screw threads. There are two main types of bolted joint designs: tension joints and shear joints.

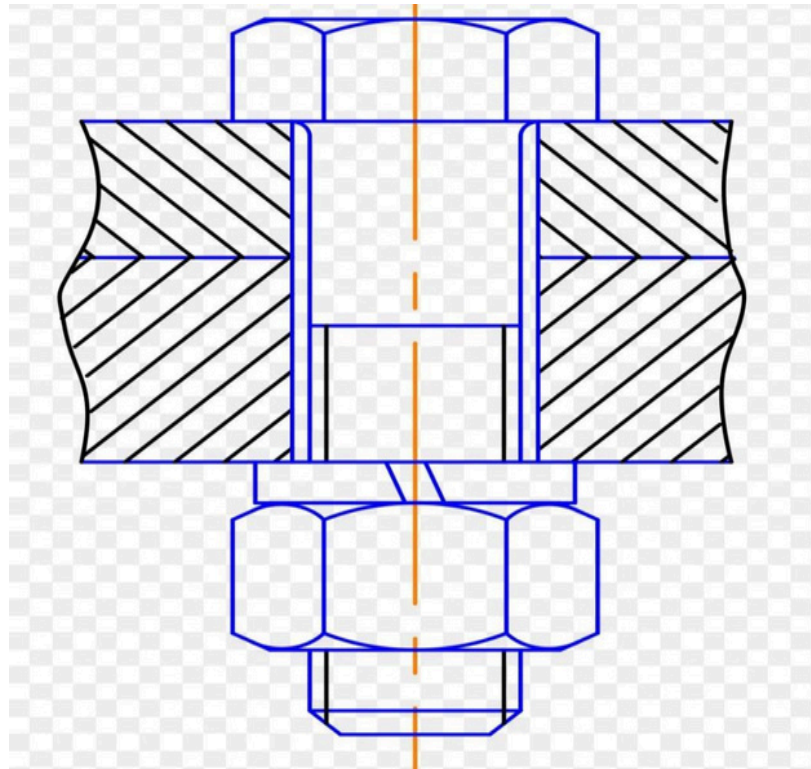


Figure 9 Bolted [8]

1.6.3.3 Factors influencing the failure of mechanical joints

- a) Types of loading
- b) Stress concentration
- c) Method of hole making
- d) Drilling induced damage
- e) Mechanical properties of fastener

1.6.3.4 Stitching

- a) Stitching is a method that increases laminar strength while at the same time preventing crack propagation.
- b) Stitching leads to better control on the handling while processing of composites.
- c) Results showed that the fracture toughness increased 30-fold when compared to the unstitched laminate.
- d) Stitching will damage the fibers and insert some holes in the composite so it will decrease the in-plane properties of composites.

1.7 Comparative study of Adhesive bonding and Mechanical joining:

1.7.1 Adhesive bonding:

- a) The heat resistance of the adhesive layer is limited. Depending on the basic material of the adhesive, temperatures for continuous stress range between approximately 120 and 300 °C.
- b) Adhesive layers and their boundary layers towards the adherents' surfaces may be damaged by environmental impacts, such as humidity, which results in a reduction of strength.
- c) In the production of bonded joints, the time required for the relevant reaction kinetics of curing is more.
- d) The growing demand for recyclability of industrial products calls for respective design-engineering measures.
- e) The availability of non-destructive test methods is rather limited.
- f) Quality control, Inspecting, repairing of complex adhesive joints is big problem.

1.7.2 Mechanical joining:

- a) These mechanical bonds are easily to repair and easily to inspect while quality control.
- b) The mechanical joints are simple in construction so production is faster.
- c) These are not affected by the environment and temperature except they undergo corrosion.
- d) These parts do not require a neat room and environment like adhesive joints.
- e) We can get a good joint by adding extra composite material, metallic reinforcements at the composite join but it will increase the overall weight of the composite structure.

1.8 Objective of using Mechanical joining over other joining process:

Mechanical joining offers many advantages compare to other fundamental joining process. The first advantages of mechanical joining are this process unique in that is primarily dependent on the structures to be joined, and only dependent on the materials which these structures are composed. No bonds need to be formed to accomplish joining, nor do any need to be broken to accomplish disassembly. Where, mechanical joining methods or technique uniquely allow simple practical disassembly without damaging the parts involved. The second advantages are uniquely allowing intentional disassembly without damaging parts involved. Where, this is very frequently essential for the purpose of portability, maintenance, services replacements, of damages parts, modification, of the assembly or structures. The third advantage is no bonds formed. Where, is the most form of mechanical fastening and some form of integral attachment permit relative intentional motion between joined parts.

Literature Review

1. G Dilli Babu, K Sivaji Babu & B Uma Maheswar Gowd (2013) et al. [9] The feed rate and cutting speed are seen to make the largest contribution to the delamination effect. Generally, the use of high cutting speed and low feed favor the minimum delamination on drilling.
2. Aravindh and Umanath (2015) et al. [10] Signal to Noise ratio analysis has been used to find the optimal drilling parameters suitable for minimal delamination which in turn improves the hole quality in drilling
3. S JAYABAL and U NATARAJAN (2011) et al. [11] The important input variables considered for the present research study is drill bit diameter, spindle speed and feed rate. The influences of all machining parameters on thrust force, torque and tool wear were analysed based on the developed mathematical model.
4. M. Ramesh, K.Palanikumar and K.Hemachandra Reddy (2014) et al. [12] During drilling of the hybrid composite laminates, different types of damages like matrix burning and cracking , fiber pullout , delamination and plastic deformation are some of the common problems occurs around the drilled hole.
5. Zhang et al. [13] have investigated the spalling, fuzzing exit sub-surface deformation defects during drilling of unidirectional as well as multi-directional CFRP laminates with HSS twist drill. Spalling and fuzzing are considered as the major exit damage mechanisms during drilling of FRP composite laminates and these damages increase with an increase in feed rate and decrease in spindle speed. Spalling at hole exit is usually a severe damage and it is bigger for UD-CFRP laminate as compared to multi-directional CFRP laminate under the same drilling conditions.
6. Khashaba et al. [14] reported that catastrophic shear failure of the composite layers has been done due to the higher feed rates and cutting temperatures resulting poor surface integrity, lower bearing strength of machined holes while GFRP composite drilling
7. Heisel et al. [15] investigated the influence of point angle, cutting parameters on cutting forces and drill hole quality i.e. frying, burr formation and delamination have been investigated while drilling of CFRP laminates. The increase in point angle increases cutting temperature resulting frying of epoxy matrix with poor quality drilled hole and severe burr formation.

8. Arul et al. [16] have utilized acoustic emission technique to improve the quality and surface integrity of the machined hole during drilling on woven glass fabric/epoxy composite laminates. The effect of cutting parameters on axial force, flank wear and their influence on the hole shrinkage was monitored and correlated with AE parameters
9. Eneyew and Ramulu et al. [17] have studied the effect of cutting parameters, cutting direction and fiber orientation on the drilled hole surface quality in the UD-CFRP composite laminate
10. Abrao et al. [18] have made an extensive literature on drilling of FRP composites. Aspects such as tool geometry, machining parameters and their influence on thrust force, torque and delamination are examined in the review.
11. Liu et al. [19] provided a review on mechanical drilling of composite materials such as FRP (CFRP, GFRP) as well as fiber metal composite (FML) laminates. This review paper also encloses with grinding drilling, Vibration-assisted twist drilling (VATD) and high-speed drilling (HSD) operations of both FRP and FML composites.

2.1 Research Gaps

- a) We will need extra machining for strenght testing.
- b) Find difficulty when we remove the fabricate composite.
- c) Drilling caused tool wear

2.2 Objective

- a) To fabricate composite with different fiber percentage (5%, 10%, and 15%).
- b) We perform FLEXURAL STRENGTH, LONGITUDINAL TENSILE STRENGTH, and COMPRESSIVE STRENGHT test on these given specimens.
- c) We will compare the strenght of these three specimens

Methodology of Work

3.1 Raw material which have been used in this process

Fibre purchased from Go Green Products.

No 225 E, First Floor, Gandhi Road, Alwarthirunagar, Chennai – 600087,

Opp –Guga marriage hall.

Landline: 044 -48513498

1. Epoxy Resin LY556 are relatively low molecular weight pre-polymers capable of being processed under a variety of conditions. Two important advantages of these over unsaturated polyester resins are: first, they can be partially cured and stored in that state, and second, they exhibit. Low shrinkage during cure.

However, the viscosity of conventional epoxy resins is higher and they are more expensive compared to polyester resins. The cured resins have high chemical, corrosion resistance, good mechanical and thermal properties, outstanding adhesion to a variety of substrates, and good and electrical properties.

Approximately 45% of the total amount of epoxy resins produced is used in protective coatings while the remaining is used in structural applications such as laminates and composites, tooling, moulding, casting, construction, adhesives

2. Hardener HY 951 with IUPAC name NNO-bis (2amino ethyl ethane-1,2diamin) has been used with the epoxy designated as HY 951. Both the epoxy and hardener were supplied by GO GREEN PRODUCT. In the present work hardener (HY951) is used. This has a viscosity of 10- 20 MPa at 25°C.

3.2 Specimen preparation

when we obtained the fibre from store. It was not clean properly so we washed the fibre with normal water to remove the dirty particle from it to get better specimen.

a) Drying and Straighten

After washing the fibre, we needed to dry it, so we left it 6 hours for sun dried. And after that we have to make them straight with help of comb.

b)Cutting

After straighten the Aloe Vera fibre we cut it size of 10 mm.

3.3 Mould Preparation

3.3.1 For Composite without Insertion

The molds have been prepared using wooden bit and plywood with dimensions of 152.4mm×152.4mm×12.7mm. Where plywood was used on top and bottom position while casting of composites. To avoid adhesion between the final product and plywood we used plastic sheet on all surfaces of mold and plywood.



Figure 10 Mould Preparation for specimen

3.3.3 Fibre preparation

The Aloe Vera fibre is obtained from aloe Vera plant, which has been collected from local sources. The extracted aloe Vera fibre were untreated but sun dried for eight hours at 28°C-30°C. When the fibre was fully dried, it was cut into the lengths of 10mm, with the help of scissors.



Figure 12 Aloe Vera Fibre



Figure 13 cutting of Fibre

3.3.4 Required Materials for Matrix

The epoxy resins and hardener are procured from Herenba instruments and Engineers, Chennai. The aloe Vera fibre based epoxy composite is fabricated using hand lay-up process.



Figure 14 Epoxy Hardener

3.3.5 Preparation before Casting

The aloe Vera fibre of 10 mm length has been mixed with matrix mixture with their respective values by simple mechanical stirring and mixtures are slowly poured in different moulds, keeping the characterization standards and view on testing condition. The releasing agent has been used on mould sheet which give easy to composites removal from the mould after curing the composites. A sliding roller has been used to remove the trapped air from the uncured composite and mould has been closed at temperature 30° C duration 24 hour. The constant load of 50 kg is applied on the mould in which the mixture of the aloe Vera, epoxy resin and hardener has been poured. After curing, the specimen has been taken out from the mould. The composite material has been cut in suitable dimensions with help of zig saw for mechanical tests as per the ASTM standards.



Figure 16 Mixing and Pouring of Fibre with Resin and Hardener

Table 2 Designation and Composition of composite

Designation	Composition
C1	Fibre length (10mm) (5 wt.%) + Epoxy (95 wt.%)
C2	Fibre length (10mm) (10 wt.%) + Epoxy (90 wt.%)
C3	Fibre length (10mm) (15 wt.%) + Epoxy (85 wt.%)



Figure 17 completed the fabrication of composite



Figure 18 After 24 hours of fabrication of composite

3.4 Specimen Preparation

3.4.1 Cutting of Specimen

The sample from the casted composite was cut in the dimension of 150mm x 25mm with the help of power cutter.

3.4.2 Filing

For smoothing of surfaces and edges of the sample filing was done. The smooth cut file was used for smoothing the corners.



Figure 19 cutting of Specimen



Figure 20 Filing of Specimen

3.5 Testing of Specimen

3.5.1 FLEXURAL STRENGTH

Flexural strength, also known as modulus of rupture, bend strength, or fracture strength, a mechanical parameter for brittle material, is defined as a material's ability to resist deformation under load. The transverse bending test is most frequently employed, in which a rod specimen having either a circular or rectangular cross-section is bent until fracture using a three point flexural test technique. The flexural strength represents the highest stress experienced within the material at its moment of rupture.

3.5.2 LONGITUDINAL TENSILE STRENGTH

Normal strength is taken because the most stress on the strain strain curve, If we tend to assume $\epsilon_f < \epsilon_m$ that is that the usual case then the fibres can fail before the matrix. Once the fibre have broken the bulk of the load that was borne by the fibres is currently transferred to the matrix. This being the case it's doable to adapt the expression for the strain on this sort of composite.

3.5.3 COMPRESSIVE STRENGTH

compressive strength test, mechanical test measuring the maximum amount of compressive load a material can bear before fracturing. The test piece, usually in the form of a cube, prism, or cylinder, is compressed between the platens of a compression-testing machine by a gradually applied load.



Future Scopes

Most of the companies are investing in R&D and are launching innovative products to gain a competitive advantage over others and to expand their global presence. Currently, one of the major trends seen in the market is the increasing use of aloe Vera extracts in sugar-free drinks, which is expected to increase the application scope for the product. Now a days the residue of Aloe Vera (i.e Aloe Vera leaf without gel) is used to make

Fibre for making composite material which has various applications in different sectors .As we can see from the above discussion ,demand of Aloe Vera is increases

(as shown in graph) as well as various future scopes Aloe Vera various fields production rate of Aloe Vera is increasing day by day ,but there is only the use of

Aloe Vera gel and leaf/peel of Aloe Vera is totally wasted . To utilize this Aloe Vera

peels Aloe Vera fiber is extracted , which further can be used in making composite material

In recent days there is various researcher is working on Aloe Vera fibreThis is completely new field of research because various experiment is to be done in future as

there is no any composite material available in the market related with Aloe Vera

fibre researcher worked only on Aloe Vera fibre and some make hybrid composite

from Aloe Vera Such as Aloe Vera –sisal composite, Aloe Vera palmary composite

,Aloe Vera and coiretc. and many more Since it is made of waste part of Aloe

Vera ,hence it is economical and ecofriendly However many work are done on Aloe

Vera in past , there is many other work can be done in future also. Some of the works

are mentioned below:-

Recently 30% Aloe Vera fibre composite is made on injection molding, so there can

be a possibility to make other composite by increasing the fiber content Many work

are done on Aloe Vera fiber composite using untreated and treated (with NaOH) Aloe

Vera , so there is possibility to get better result using other chemicals treatment , with

different concentration .

Conclusion

Demand of polymer composite material increases in last few decades. It is used in many structural and non-structural application. A number of products has been started to manufacture using these materials. Therefore, development of new product with intricate shape requires joining of individual component in an assembly and joining of composites is challenging task.

1 Material properties of thermosets and thermoplastics are different. Thermosets cannot melt again after curing. Therefore, it is not suited for fusion bonding.

Adhesive joining and mechanical joining can be used for thermoset based composites. Surface energy, surface roughness and chemical property of adherend surface plays important role in adhesive joining.

2 Thermoplastic matrix composite cannot be used for adhesive joining. The surface energy available at the surface of composite is very low which provide weak bonding strength between adherend. Corona discharge treatment, plasma treatment or ion treatment are utilised to improve the surface energy of thermoplastic matrix composite.

3 Sustainability: Aloe vera fiber composites are derived from a renewable resource, aligning with the global emphasis on sustainability and environmental responsibility.

Biodegradability: A significant advantage of aloe vera fiber composites is their biodegradability, offering a solution to the environmental challenges associated with non-biodegradable materials.

4 Fusion bonding is used for thermoplastic matrix composites. Thermoplastic can be melted again after solidification. This property of thermoplastic makes it successful for fusion bonding. Amount of heat depends on amorphous and crystalline phase present in thermoplastic. Chemical bonding, chain entanglement and chain diffusion is basic mechanism of bond formation in fusion bonding.

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